

Mohamad Awad

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Civil engineering student at Toronto Metropolitan University. I build software for pavement inspection, public data and document automation, with an emphasis on traceable results and clear limitations.

Education

Bachelor of Engineering, Civil Engineering | Toronto Metropolitan University

September 2024 to June 2028 (expected) | Toronto, Ontario

Experience

Data Engineer, work placement | Toronto smart-city startup

May to July 2026

- Developed Python ingestors for six City of Toronto mobility datasets, with data contracts, tests and documentation; submitted the work for technical review.
- Built a signal engine over eight years of Toronto 311 data, investigating category renames and missing-month artifacts before reporting trends.

Document Automation Workstream Lead | Pop-Up Chapel Co. (Riipen)

April to May 2026

- Led document automation within a student consulting engagement: a prototype generates eight branded documents in about ten seconds per booking, against an estimated eight hours of manual preparation.
- Built the Python generation pipeline and a Next.js companion tool; mapped client communication touchpoints and identified missing messages.

Fitness Centre Desk Supervisor | Toronto Metropolitan University

May 2025 to July 2026

- Coordinated daily floor operations for a team of 35 and inspected 30 or more pieces of equipment per shift, filing maintenance requests when needed.

Project Coordinator | BrandEQ

June to August 2024

- Tracked timelines and deliverables across six client accounts and prepared weekly client progress updates.

Selected projects

PaveScan AI | Independent project

2026

- Built a dashcam inspection pipeline combining computer vision, GPS-linked findings, a density-based condition estimate and PDF reporting.
- The bundled Toronto sample covers 3,571 m of scored road and 693 frames. Its 385 model findings include 275 possible shadows; outputs are review aids, not certified field inspections.
- Reworked scoring around fixed sample units after finding that the retired scorer changed with reporting-segment length.

LocalFlow | Personal software project

2026

- Built local voice dictation with Whisper and a local language model, integrating a hotkey-to-text workflow with GPU and CPU processing paths.

Engineering coursework: requirements, sketches and CAD concepts for a mobile storage cart and walking cane. Design work only; no physical fabrication claimed.

Technical skills

Engineering: AutoCAD, Civil 3D, Revit, SolidWorks, Microsoft Project, Excel.

Software and data: Python, TypeScript, React, Next.js, PostgreSQL, ETL, Git, pytest, Playwright.

Computer vision: PyTorch, YOLO11, OpenCV, ONNX Runtime, segmentation and model evaluation.